Vishal Singh

vishalsingh17-web2 | vishal-singh-1703 | My Portfolio | vishal.workcontact@gmail.com | +917084324572 | LeetCode

Work Experience

Android Developer Oct 2022 - present

AFFINSYS AI PVT LTD (Internship + Full Time)

- Led development of **mobile apps from scratch** using React Native in Typescript.
- Worked on project **autogen**, a backend driven form builder using JSON rendering, which **reduced** the overall **development time** by 60%.
- Wrote **payment processors** using **Strategy pattern**, handled retries and failures using state observer.
- Built **CI/CD pipeline** using Github Actions/Fastlane for android application, enhancing build stability.
- Worked on **Native/Turbo modules** to deal with native APIs.
- Added security checks to handle **Frida Injections** and **SSL Pinning** to prevent reverse engineering.
- Added **Internationalization** support for **multiple languages** in the app.
- Worked closely with **Vision camera** and **ML Kit** to implement **QR code scanning** and **face detection** features.
- Added **OTA updates** and **Firebase Crashlytics/Sentry** for crash reporting.
- Implemented **deep linking, notification system** for user engagement.

Key Learnings – React | React Native | Redux/RTK Query | Fastlane | Reducers/Context API | NPM Packages | Push notification | Native modules | CI/CD | Typescript/Javascript | JEST | Webviews/PWA

Flutter Developer Jan 2022 - April 2022

TREEVED (Internship)

- Debugged, managed, and created new screens and features for the app
- Learned **state management** in Flutter using **Provider and Bloc package** to effectively manage app resources
- Developed a Flutter/Dart-based **chrome extension**
- Added **Method channels** to call the native APIs on android side.
- Made API calls using the **repository pattern** with proper implementation of abstract classes

Key Learnings – Flutter bloc/Provider | Chrome Extension | Method channels | Material design | Firebase

Flutter Developer Nov 2021 - Jan 2022

CRUV. (Internship)

- Implemented **pixel-perfect UI** designs
- Utilized **isolates** to enhance app performance
- Employed **isolates for calling APIs** in separate memory threads
- Managed **heavy background tasks** such as video downloading and running complex algorithms

Key Learnings – Flutter/Dart | UI Design/Building | Isolates

Projects

Fall Detection

- Project based on ML/DL, which detects fall, using smart phone sensors.
- Created our own dataset using flutter app and on-board sensors of smartphone (Accelerometer, gyroscope and magnetometer)
- Analyzed it with different machine learning and deep learning algorithms, Got accuracy of 99.25% using Random forest
- Flutter app to predict the fall and other daily life activities (Walking, sitting, standing, running).

IoT Smart Farming

- Implemented it with using Node-MCU, flutter and firebase.
- Collected data from sensors (DHT11 and Soil moisture sensor), and stored them on RLTD of firebase
- fetched the data to our app based on flutter, also generated alerts and controlled the motor-pump using our app.

Chat Application (Flutter)

- Project based on Flutter/dart with Google authentication, Supports text-based messages including emojis, images and multi-line text.

Education

Bachelor of Technology, (ECE) Vellore institute of technology, Vellore (GPA: 9.09/10.0)
HSC/Intermediate (Class 12th) Shah Faiz Public School (Percent: 91.6 %)
HSC (Class 10th) Shah Faiz Public School (GPA: 9.6/10.0)

Skills

Frameworks	React, React native, Flutter, Jetpack compose, Android
Programming Skills	Dart, Python, C, C++, Javascript, Typescript
Essentials	Git and GitHub/Gitlab, Postman, Swagger, Jupyter-Notebook